

Gaming Dataset Analysis

Exploring game ratings by release
date

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Overview

Business Problem

We need to determine whether video games released between 2001-2010 have significantly higher ratings than those released between 2011-2020. Understanding this trend will help us make informed decisions about game design, market preferences, and potential investment in specific genres or time periods.

Dataset Overview

- Source: Kaggle (The data has been collected from a game log/recording site called backloggd.)
 - Simple data scraping tools have been used such as Beautiful Soup*.
- 1512 Samples, 13 Features
- Features Used:
 - Title
 - Release Year (created column)
 - Release Date
 - Team
 - Rating
 - Genre

*Beautiful Soup is a web page scraping tool package available in python which simplifies HTML and XML parsing.

Overview

Business Question

Do games released between 2001-2010 have a better player ratings compared to games released in 2011-2020 from backloggd.com.

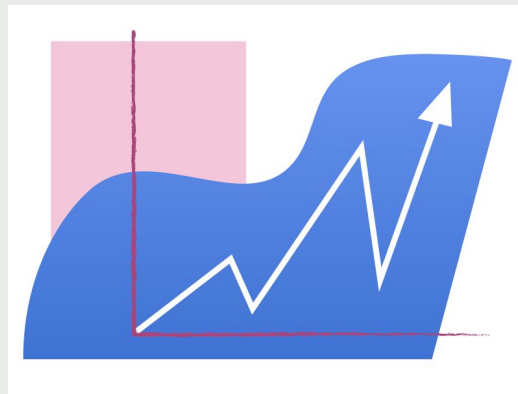
Hypothesis

Null Hypothesis:

There is no significant difference in the average ratings of games released between 2001-2010 and 2011-2020

Alternative Hypothesis:

Games released in 2001-2010 have significantly higher ratings than those in 2011-2020.



Methods Results and Findings

01

Methods

- Data was grouped and sampled
- Alpha 0.05
- Used a 1 sided t-test
 - Greater sided
- Observe P-Value

02

Results

T-Test: 2.521243380534123
P-value: 0.00624121392540133

03

Findings

P-Value is not greater than our Alpha.

In favour of the alternative.

Data Cleaning and Wrangling

```
df.dtypes
```

```
Title           object
Release Date    object
Team            object
Rating          float64
Times Listed    object
Number of Reviews object
Genres          object
Summary         object
Reviews         object
Plays           object
Playing         object
Backlogs        object
Wishlist        object
dtype: object
```

```
df.isnull().sum()
```

```
Title           0
Release Date     3
Team             1
Rating          13
Times Listed     0
Number of Reviews 0
Genres           0
Summary         1
Reviews         0
Plays           0
Playing         0
Backlogs        0
Wishlist        0
Duplicate_Status 0
dtype: int64
```

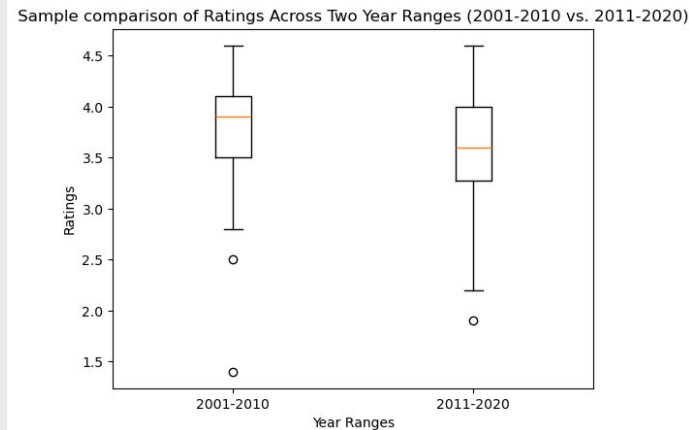
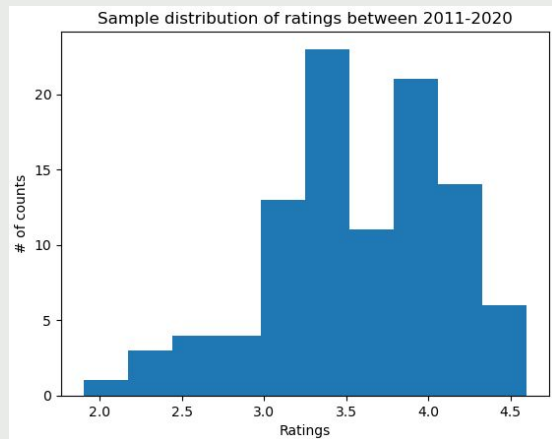
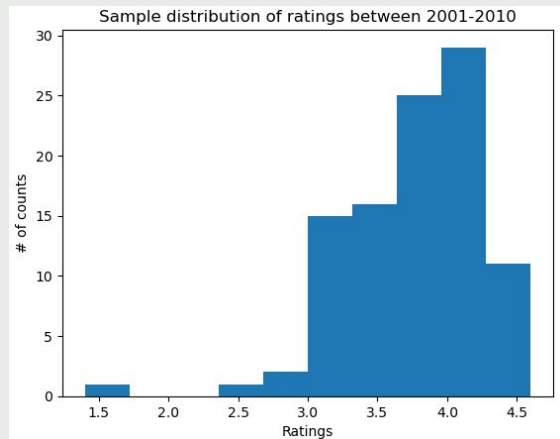
```
# Find rows where 'Genres' is an empty list or an empty string
missing_genres = df[df['Genres'].isin(['[]', ''])]
missing_genres_rows = missing_genres.index.tolist()
# Display the indices of such rows
print(missing_genres_rows)
```

```
[481, 896, 1061]
```

```
df["Duplicate_Status"].value_counts() #Unique vs Duplicate check
```

```
Duplicate_Status
Unique      1099
Duplicate    413
Name: count, dtype: int64
```

Exploratory Data Analysis

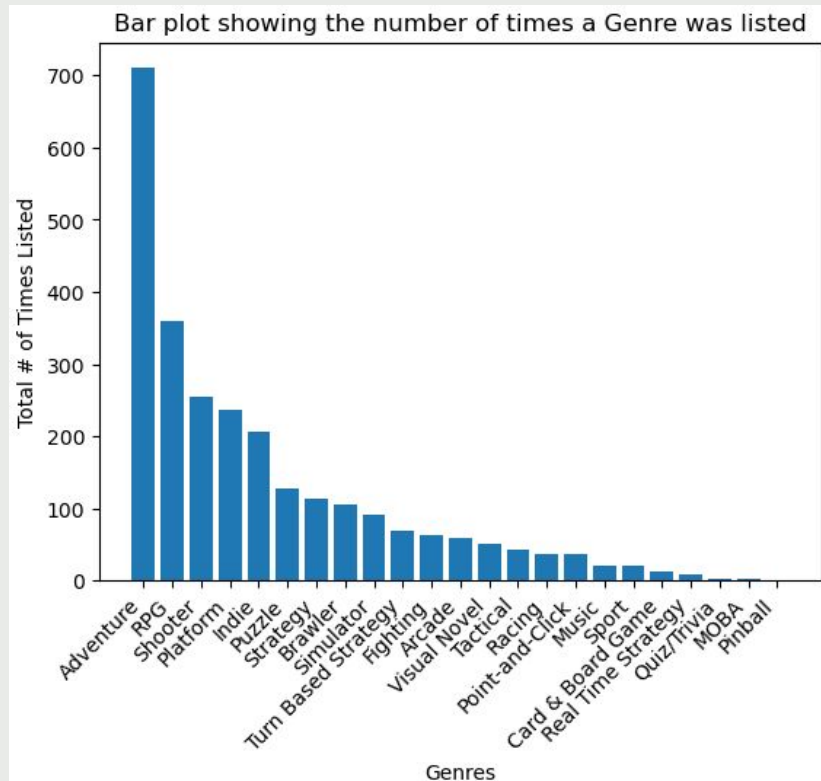


Conclusion

- Games created between 2001-2010 have better player rated games compared to releases in 2011-2020
- Companies should consider older concepts in games compared to newer games.



Bonus



```
# Print the top teams
```

```
print(team_counts)
```

```
Nintendo      170
```

```
Capcom        62
```

```
Square Enix   56
```

```
Sega          44
```

```
Electronic Arts 42
```

```
...
```

```
Warhorse Studios 1
```

```
Cold Symmetry    1
```

```
Playstack        1
```

```
ThatGameCompany  1
```

```
TT Fusion         1
```

```
Name: count, Length: 662, dtype: int64
```



Future Improvements

- Fill null genre values using previous iterations of the game
- Fill null rating values with mean value from same genre

